Memory efficiency and resource-rational encoding in human sentence processing

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What are working memory (WM) resource constraints? How can we represent and manipulate them at the computational level? How do they shape the representational space in sentence processing? Compared to previous studies that focus on memory retrieval mechanism[1-2], in this work, we argue that greater WM resources result in lower representational uncertainty in memory encodings of linguistic inputs (Fig 1a)[3-4]. We propose a new paradigm to represent, evaluate, and manipulate WM constraints in Transformer-based language models (Fig 1b). **Findings:** explicit WM constraints on encoding precision can help the model better align with human reading times (RTs) (Fig 2), and importantly result in a more compressed representational space in the sense that their next-word predictions become more similar across different contextual inputs (Fig 3).

Models. We modify the architecture of GPT-2 by injecting noise into the self-attention value vectors. As in Fig 1b, the noisy value vector $\tilde{\mathbf{v}}_{qk}$ at key position k given query position q is sampled from a Gaussian distribution, whose precision τ exponentially decays as a function of key-query distance ($q-k$) and is controlled by the decay rate θ. The models are trained on `wikitext` with a hybrid loss function: in addition to a cross-entropy objective, the loss function also includes a memory capacity term, defined as the precision τ averaged over all key and query tokens across all attention heads. This memory component penalizes models with higher precision, with λ representing the penalty strength such that higher λ corresponds to a stronger memory capacity constraint.

Memory precision decay rates learned by the models are shown in Fig 1c. First, when there is no memory capacity constraint ($\lambda = 0$), all heads learn a decay rate close to 0. Intuitively, this suggests that it is useful to use all available information from the preceding context to maximize the accuracy of next-word prediction, consistent with the memory-surprisal trade-off[5]. Then, with lower-to-mid memory constraint ($0 < \lambda \leq 0.01$), the precision decay first begins among attention heads at lower layers, which often take charge of the processing of morphosyntactic features where information dependency is more local. As memory constraint gets stronger ($\lambda > 0.01$), the decay starts to spread to higher layers, which often process longer discourse context.

Alignment with human RT. The predictive power of model-generated surprisals on RTs, quantified as ΔlogLik between regressions with and without surprisal terms[6-7], is evaluated on 3 English reading corpora: Provo[8](eye-tracking), Natural Stories SPR[9], Natural Stories Maze[10]. ***Result.*** As in Fig. 2, explicit WM constraint can help the model better align with human RTs, except for the Maze corpus, which is an experimental paradigm that typically requires higher memory load.

Representational space of next-word prediction. As shown in Fig 3a, we first construct a set of 1000 items in English, all in the form of `Det N1 Prep Det N2 Adv`. The animacy of `N1` is manipulated such that 500 items have animate `N1` and the other 500 have inanimate `N1`. Then, the representational space of next-word prediction is constructed based on the log probabilities over the vocab generated at the last token of each item. ***Result.*** As shown in Fig 3bc, with no memory constraint ($\lambda = 0$), the model makes sharper next-word prediction in the sense that each item yields a more distinct distribution over possible continuations. With stronger memory constraints, the representational space is more compressed, and next-word predictions become more similar across items, both overall and within each animacy category. Moreover, as λ increases, separated prediction clusters start to emerge. Overall, this result suggests that higher memory constraints may have encouraged the model to pursue more categorical prediction strategies, shifting away from fine-grained semantic distinctions towards higher-level generalization[11-12].

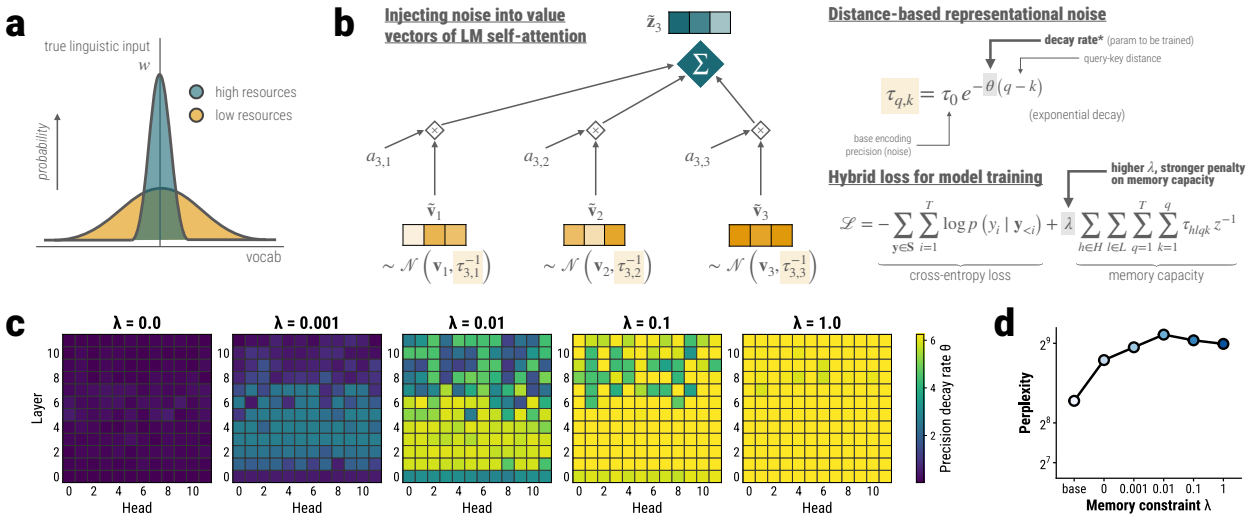

Figure 1: **a.** More memory resources result in lower representational noise for encoded linguistic inputs. **b.** Modeling framework: injecting noise into transformer LMs such that self-attention value vectors **v** are sampled from Gaussian distribution, whose precision is a function of query-key distance controlled by decay rate θ; models are trained on hybrid loss, regularizing the encoding precision as controlled by λ. **c.** Precision decay rate θ learned by models across attention heads per layer. **d.** Model perplexity on RT corpora.

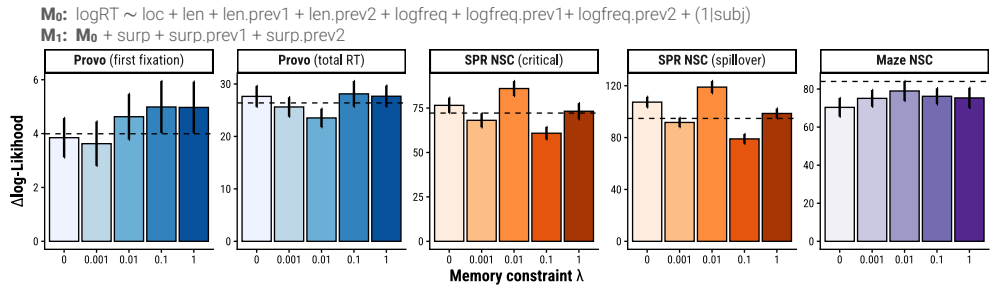

Figure 2: The predictive power (ΔlogLik) of word surprisals on RTs across models with varying WM constraints λ. Each bar is the mean ΔlogLik across 10 held-out folds with standard error. Dashed lines are ΔlogLik of the baseline GPT-2 model.

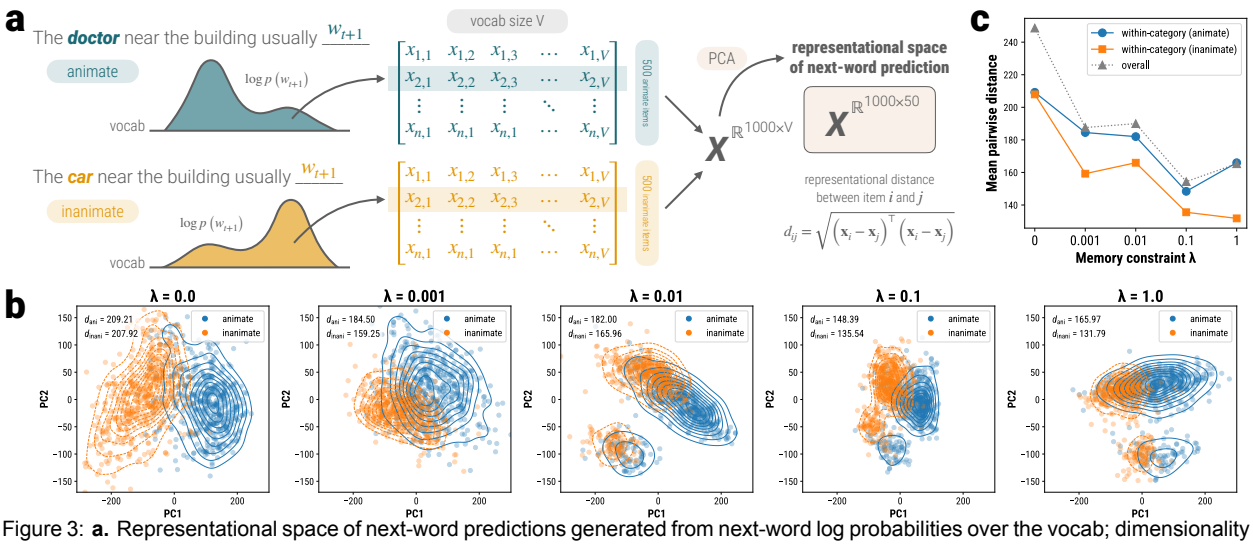

Figure 3: **a.** Representational space of next-word predictions generated from next-word log probabilities over the vocab; dimensionality is reduced using PCA with 50 PCs. **b.** Representational space across models with varying memory constraints, visualized from the first 2 PCs. **c.** Euclidean distance (based on 50 PCs) averaged over all pairs of items and within each animacy category.

References. [1] de Varda & Marelli (2024) *CMCL*; [2] Clark et al. (2025) *COLING*; [3] Bays et al. (2024) *Nature Human Behav*; [4] Xu & Futrell (2026) *JML*; [5] Hahn et al. (2021) *Psych Rev*; [6] Wilcox et al. (2023) *TACL*; [7] Xu et al. (2023) *EMNLP Findings*; [8] Luke and Christianson (2018) *Behav Rsrch Methods*; [9] Futrell et al. (2021) *Lang Rsrces and Eval*; [10] Veronica & Roger (2023) *Glossa*; [11] Bates & Jacobs (2020) *Psych Rev*; [12] Oh & Linzen (2025) *arXiv*.